



RAGIB AHNAF NEHAL

Software Engineer (Machine Learning, Flutter & Full-Stack Developer)

+880 1560-045148

ran.ragibahnafnehal2@gmail.com

linkedin.com/in/ragib-ahnaf-nehal

Dhaka, Bangladesh

github.com/ranehal

https://ranehal.vercel.app

ABOUT ME

Computer Science & Engineering graduate from United International University (CGPA 3.44) specializing in Machine Learning, Federated Learning, Mobile App Development (Flutter/Dart), and Full-Stack Systems. Proven experience architecting offline-first mobile OLAP engines (ShopSathi with embedded DuckDB querying 110k+ SKUs in <50ms), distributed TinyML clusters achieving 96% accuracy on edge medical devices, and high-throughput automated data ingestion pipelines. Certified in Cisco DevNet, Cisco Cybersecurity, and Red Hat System Administration (RH124/RH134).

KEY TECHNICAL PROJECTS

2025 – Present, Lead Dev

ShopSathi Engine

github.com/ranehal/ShopSathi

Lead Mobile Engineer & Architect (Flutter, DuckDB OLAP & AI Cascade)

- Architected an offline-first grocery price intelligence engine across 8 Bangladeshi supermarket chains (110k+ SKUs, 15.7M price points).
- Engineered an embedded vectorized DuckDB OLAP engine in Dart, executing analytical SQL on local Parquet files in <50ms.
- Integrated multi-model GenAI cascade (Gemini Flash Lite, Groq, OpenRouter) with mathematical fallback and PBKDF2/AES-256-GCM encryption.

2025 – 2026, Thesis Lead

UIU Research Lab

github.com/ranehal/FYDP

Lead Machine Learning Researcher — FedTier (Federated TinyML)

- Engineered a Federated TinyML framework for distributed medical imaging diagnostics across a 5-node non-IID client cluster.
- Benchmarked EfficientNet-B0 and MobileNetV2/V3 under FedAvg/FedProx; achieved 96.0% accuracy, 92.7% compression, and 70% less payload.

2024 – 2025, Open-Source

GroceryGOD Pipeline

ranehal.github.io/GroceryGOD

Data & Automation Engineer (Python, Async I/O & Scraping)

- Built a high-throughput parallelized automated data ingestion pipeline scraping 116k+ grocery SKUs across 4 major retail chains.
- Implemented schema validation, anomaly filtering, automated GitHub Pages dataset publishing, and real-time Telegram Bot alerting.

2024, Full-Stack Lead

FoodPANDA Analytics

ranehal.github.io/FoodPANDA

Full-Stack Web Engineer (Next.js, WebAssembly & Parquet)

- Built live analytics dashboard consuming REST/GraphQL APIs with OAuth2 lifecycle management and exponential backoff retry.
- Integrated in-browser WebAssembly Parquet decoding (hyparquet), compressing 81.7MB raw JSON to 4.7MB Parquet on GitHub Pages.

2024, Urban Logistics

LEBUtracker Transit

ranehal.github.io/LEBUtracker

Frontend & Telematics Engineer (JavaScript, PWA & Geolocation)

- Engineered a live transit tracker for Dhaka feeder buses and university shuttles with real-time routing and client-side caching.
- Implemented browser Geolocation broadcasting, dynamic Leaflet map rendering, Bangla UI localization, and PWA offline support.

TECHNICAL SKILLS

Languages : Python 3.x, Dart, JavaScript (ES6+), TypeScript, C, C++, SQL, Bash Scripting, HTML5, CSS3

Development Skills : Flutter 3.x, Riverpod, Bloc, Android SDK, React 19, Next.js 16, React Native, Node.js, Express, FastAPI, REST & GraphQL APIs, WebSockets, Embedded DuckDB (OLAP), Apache Parquet, SQLite, Hive, Chart.js, Clean Architecture, CI/CD (GitHub Actions)

Machine Learning & AI : TensorFlow, Keras, PyTorch, Scikit-learn, Federated Learning (FedAvg, FedProx, FedBN), TinyML, MobileNetV2/V3, EfficientNet, LLM Integration (Gemini, Groq, OpenRouter), XGBoost, LightGBM, CatBoost, Optuna Hyperparameter Optimization

Tools, Security & Systems : Git, GitHub, VS Code, Android Studio, Kaggle, Docker, Postman, Linux (RHEL, Ubuntu), Bash CLI, AES-256-GCM, PBKDF2, Socket Debugging, Wi-Fi Protocols (WPS), Red Hat System Admin (RH124/RH134), Cisco Networking (CCNA/DevNet)

ACHIEVEMENTS & CERTIFICATIONS

Verified Professional Certifications (Cisco & Red Hat):

- Cisco Certified DevNet Associate (Cisco, Mar 2026) • Junior Cybersecurity Analyst & Intro to Cybersecurity (Cisco, 2026)
- Red Hat Certified System Administration (RH124 & RH134) (Red Hat, Jun 2025) • CCNA: Intro to Networks (Cisco, 2024)

Open-Source Bug Triage & Network Protocol Investigations:

Active contributor investigating real-world network resilience: diagnosed SSL/TLS connection pool exhaustion (WinError 10060), GraphQL socket timeouts (ETIMEDOUT), and proxy rotation resilience in rdavydov/Twitch-Channel-Points-Miner-v2 and Zaarg/DropBot; contributed to fulvius31/OneShot (WPS auditing) and beradeep/whitehole.

GitHub Portfolio & Open Source Footprint:

<https://github.com/ranehal> — 37 public repositories and 218+ star engagements across cybersecurity tooling, ML pipelines, and web utilities.

EDUCATION

2022 – 2026

United International University

Bachelor of Science in Computer Science & Engineering (B.Sc. CSE)

CGPA: 3.44 / 4.00 | Focus: Intelligent Systems, Federated Learning & Software Engineering